Objective

To implement and demonstrate Multilevel Inheritance in Java, where the Puppy class inherits from Dog, and Dog inherits from Animal.

Steps for Execution

1.Create a Java project and create a file named Main.java.

2.Define the Animal class with the eat() method.

3.Define the Dog class extending Animal.

4.Define the Puppy class extending Dog.

5.Create a Puppy object in the main() method.

6.Call the inherited and own methods.

7.Compile and run the program.

8.Observe the output.

Algorithm

1.Start the program.

2.Define the Animal class with eat().

3.Define Dog extending Animal with bark().

4.Define Puppy extending Dog with weep().

5.Create a Puppy object.

6.Call eat(), bark(), and weep().

7.Display the output.

8.Stop the program.

UML Class Diagram

```
┌──────────────────────────┐
│      Animal      │
├──────────────────────────┤
│ + eat() : void    │
```

```
└───────────────▲───────────────┘
      │
      │ extends
      │
┌───────────────┴───────────────┐
│      Dog       │
├───────────────────────────────┤
│ + bark() : void      │
└───────────────▲───────────────┘
      │
      │ extends
      │
┌───────────────┴───────────────┐
│     Puppy       │
├───────────────────────────────┤
│ + weep() : void     │
└───────────────────────────────┘
```

Relationships

- Dog extends Animal
- Puppy extends Dog
- Puppy indirectly inherits the eat() method from Animal.
- Puppy inherits the bark() method from Dog.
- Puppy has its own weep() method.

CODE:

```
class Animal {
    void eat() {
        System.out.println("Animal eats food");
    }
```

```
}

class Dog extends Animal {
    void bark() {
        System.out.println("Dog barks");
    }
}

class Puppy extends Dog {
    void weep() {
        System.out.println("Puppy weeps");
    }
}

public class Main {
    public static void main(String[] args) {
        Puppy p = new Puppy();

        p.eat();   // inherited from Animal
        p.bark();  // inherited from Dog
        p.weep();  // method of Puppy
    }
}
```

Output

Animal eats food

Dog barks

Puppy weeps

Result

The program successfully demonstrates Multilevel Inheritance in Java. The Puppy class inherits the bark() method from Dog and indirectly inherits the eat() method from Animal, while also having its own weep() method.